

COMPUTER SCIENCE & IT

DIGITAL LOGIC



Lecture No: 03

Miscellaneous Topics



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Recap of Previous Lecture



- Number System
-
-



Topics to be Covered

Signed No. representation



[Question]

9's and 10's compliment of 123.123 is

(a) 876.876, 877.876 ✗

✓ (b) 876.876, 876.877

(c) 987.987, 988.987

(d) 987.987, 987.988

↓ 9's
876.876

↓ 10's
876.877

[Question]

1's and 2's compliment of 110.110 will be :

(a) 001.001, 010.001 ~~X~~

(b) 001.001, 011.001

☒ (c) 001.001, 001.010

(d) 011.011, 011.010

$\downarrow 1's$
 001.001
 $\downarrow 2's$
 001.010

[Question]



A no. N_1 is given as $N_1 = (1010111010000000)_2$ the 2's compliment of N_1 is N_2 . Total no. of 1's in N_2 is 4.

$$N_2 = (0101000110000000)_2$$

[Question]

2's compliment of no. N_1 is given as $(110.011)_2$ then no. N_1 is

☒ (a) $(001.101)_2$

(b) $(001.100)_2$

(c) $(010.100)_2$

(d) $(001.011)_2$

$\downarrow$ 2's
 $N_1 = (001.101)_2$

[Signed No. Representation]



Sign magnitude

1's complement

2's complement

- The representation of +ve number is same in all of the above representation.

[Sign Magnitude Method]

4-bit

0 0 0 0 (+0)	1 0 0 0 (-0)
0 0 0 1 (+1)	1 0 0 1 (-1)
0 0 1 0 (+2)	1 0 1 0 (-2)
0 0 1 1 (+3)	1 0 1 1 (-3)
0 1 0 0 (+4)	1 1 0 0 (-4)
0 1 0 1 (+5)	1 1 0 1 (-5)
0 1 1 0 (+6)	1 1 1 0 (-6)
0 1 1 1 (+7)	1 1 1 1 (-7)



- Total distinct numbers that can be represented using n -bit in sign magnitude method $= (2^n - 1)$

- Range of the numbers that can be represented using n -bit

$$-\left[2^{n-1} - 1\right] \quad \text{to} \quad +\left[2^{n-1} - 1\right]$$

[1's Complement Method]

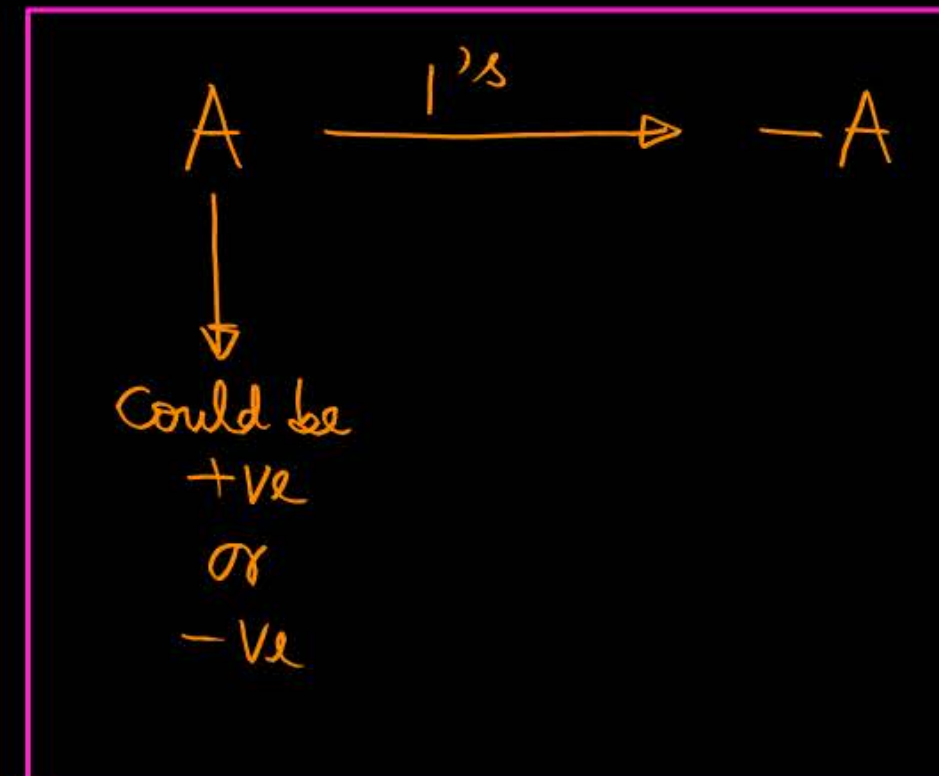


4-bit

0 000	(+0)	$\xrightarrow{1's}$	1111	(-0)
0 001	(+1)	$\xrightarrow{1's}$	1110	(-1)
0 010	(+2)	$\xrightarrow{1's}$	1101	(-2)
0 011	(+3)	$\xrightarrow{1's}$	1100	(-3)
0 100	(+4)	$\xrightarrow{1's}$	1011	(-4)
0 101	(+5)	$\xrightarrow{1's}$	1010	(-5)
0 110	(+6)	$\xrightarrow{1's}$	1001	(-6)
0 111	(+7)	$\xrightarrow{1's}$	1000	(-7)

$$\begin{array}{ccc} 10111 & \xrightarrow{1's} & (01000) \\ (-8) & & +8 \end{array}$$

$$\begin{array}{ccc} 1011 & \xrightarrow{1's} & 0100 \\ (-4) & & (+4) \quad -(-4) \end{array}$$



- Total no. of distinct numbers that can be represented using n -bit in 1's complement representation $= (2^n - 1)$

- Range of the numbers that can be represented using n -bit:

$$- [2^{n-1} - 1] \quad \text{to} \quad + [2^{n-1} - 1]$$

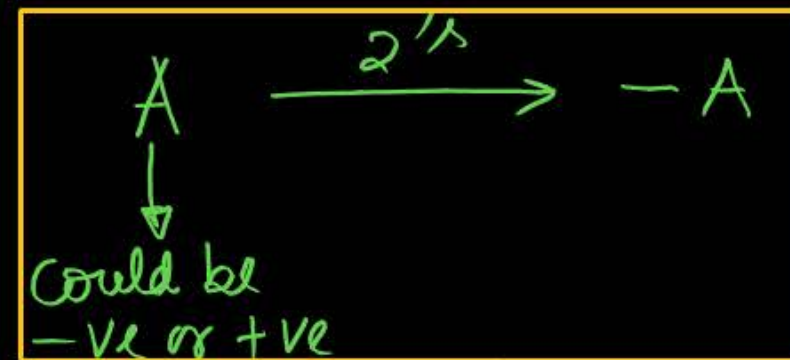
[2's Compliment Method]



4-bit

0 0 0 0	(+0)	$\xrightarrow{2's}$	0 0 0 0	(-0)
0 0 0 1	(+1)	$\xrightarrow{2's}$	1 1 1 1	(-1)
0 0 1 0	(+2)	$\xrightarrow{2's}$	1 1 1 0	(-2)
0 0 1 1	(+3)	$\xrightarrow{2's}$	1 1 0 1	(-3)
0 1 0 0	(+4)	$\xrightarrow{2's}$	1 1 0 0	(-4)
0 1 0 1	(+5)	$\xrightarrow{2's}$	1 0 1 1	(-5)
0 1 1 0	(+6)	$\xrightarrow{2's}$	1 0 1 0	(-6)
0 1 1 1	(+7)	$\xrightarrow{2's}$	1 0 0 1	(-7)

$$(1000)_2 = -8 \times 1 + 0 + 0 + 0 = (-8)$$



$$1110 \xrightarrow{2's} 0010$$

(-2) $\quad$ (+2) $\quad$ -(-2)

$$= -8 + 4 + 2 + 0$$

$$= (-2)_{10}$$

$$(1011)_2 = -8 + 0 + 2 + 1$$

$$= (-5)_{10}$$

Total no. of distinct numbers that can be represented using n -bit in 2's
Compliment representation = 2^n

- Range of the numbers using n -bit

$$- [2^{n-1}] \quad + [2^{n-1} - 1]$$

Imp Discussion regarding 2's complement representation

Note: Copying MSB to the left of MSB any no. of times will not change the number in 2's complement representation.

$$1010 = (-6)_{10} = -8 + 0 + 2 + 0$$

$$1\ 1010 = (-6)_{10} = -16 + 8 + 0 + 2 + 0 \xrightarrow{2's} (00110)$$

$$11\ 1010 = (-6)_{10} = -32 + 16 + 8 + 0 + 2 + 0$$

$$0100 = +4$$

$$00100 = +4$$

• $(1101)_2$

represent this no. in 8-bit $\longrightarrow (1111\ 1\underline{1}01)_2 = (101)_2 = -4 + 1 = (-3)_{10}$

$$(1111\ 1\underline{1}0101)_2 = (10101)_2 = (-11)_{10}$$

- Imp Points :

$$1011 = -5$$

$$\begin{array}{ccccccc} & \swarrow & \swarrow & \swarrow & \swarrow & & \\ 1 & 0 & 1 & 1 & 0 & = & -10 = 2 \times (-5) \end{array}$$

$$101100 = -20 = 2 \times 2 \times (-5)$$

$$011 = +3$$

$$0110 = +6 = 2 \times 3$$

$$01100 = +12 = 2 \times 2 \times 3$$

[Addition and Subtraction in 2's complement signed no. representation]



Lets understand with example :

$$A - B = A + (-B)$$

$$B \xrightarrow{2's} -B$$

- $A = 1011 = -5$

$$B = 0100 = +4$$

$$\begin{array}{r} + \\ 1011 \\ \hline 1111 = (-1)_{10} \end{array}$$

- $A = 1100 = (-4)$

$$B = 0111 = +7$$

$$\begin{array}{r} \leftarrow \textcircled{1} 0011 \rightarrow \text{Result} = (0011)_2 = (+3) \\ \text{discard} \end{array}$$

- $A = 1100 = -4$ $-(8) \text{ to } (7)$

$$B = 1110 = -2$$

$$\begin{array}{r} + \text{Carry} \uparrow \\ \text{Cin} \uparrow \\ \leftarrow \textcircled{1} 1010 \rightarrow \text{Result} = (1010)_2 = (-6)_{10} \\ \text{discard} \end{array}$$

- $A = 0010 = +2$

$$B = 0100 = +4$$

$$\begin{array}{r} + \\ 0110 \rightarrow \text{Result} = (0110)_2 = (+6)_{10} \end{array}$$

- $A = 1011 = -5$

$$B = 1100 = -4$$

$$\begin{array}{r} \text{Cout} \nearrow \text{Cin} \\ \text{discarded} \quad \text{①} \quad 0111 \end{array} \Rightarrow \text{Result} = (0111)_2 = (+7)$$

- $A = 0101 = +5$

$$B = 0011 = +3$$

$$\begin{array}{r} \text{Cout} \nearrow \text{Cin} \\ 1000 \end{array} \rightarrow \text{Result} = (1000)_2 = (-8)_{10}$$

- $A = 0110 = +6$

$$B = 0100 = +4$$

$$\begin{array}{r} 1010 \end{array} \rightarrow \text{Result} = 1010 = (-6)_{10}$$



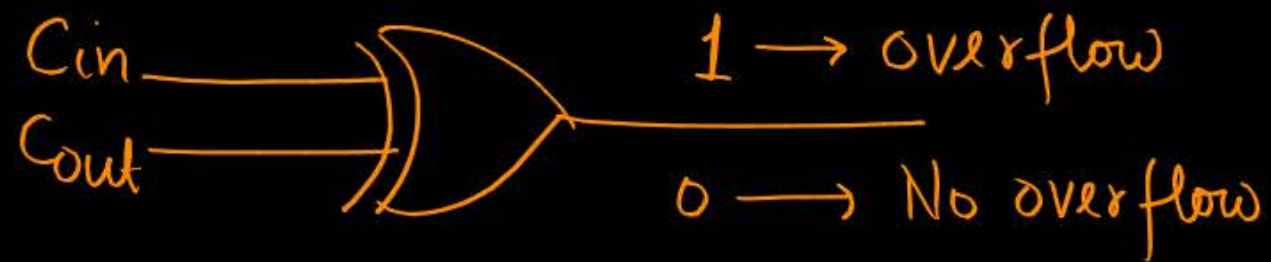
[Condition of Overflow]

$x \rightarrow$ sign bit of A

$y \rightarrow$ sign bit of B

$z \rightarrow$ sign bit of Result of (A+B)

$$\text{Overflow}(x, y, z) = \overline{x} \cdot \overline{y} \cdot z + x \cdot y \cdot \overline{z}$$



- $A = 1010 = 10$
 $B = 0110 = 6$

} unsigned numbers

$$(A - B) = A + 2^{\text{'s}} \text{ complement of } B$$

$$\begin{array}{r}
 1010 \\
 + 1010 \\
 \hline
 \textcircled{1} 0100 \\
 \hline
 \end{array}
 = + (0100)$$

Result is +ve = + (4)

- $B - A \rightarrow$

$$\begin{array}{r}
 0110 \quad B \\
 + 0110 \quad (2^{\text{'s}} \text{ complement of } A) \\
 \hline
 1100 \\
 \hline
 \end{array}
 = - (2^{\text{'s}} \text{ complement of } 1100)$$

no carry generation $\rightarrow$ Result is -ve = - (0100) = (-4)

$$\begin{array}{r}
 A = 1101 = 13 \\
 B = 0010 = 2
 \end{array}$$

$$\begin{array}{r}
 A - B \rightarrow \begin{array}{r} 1101 \\ + 1110 \\ \hline \end{array}$$

A
2's complement of B

$$\begin{array}{r}
 \textcircled{1} 1011 \\
 \hline
 \end{array}
 = + (1011)$$

Result is +ve = + (11)₁₀

$$\begin{array}{r}
 B - A = \begin{array}{r} 0010 \\ + 0011 \\ \hline \end{array}$$

B
2's complement of A

$$\begin{array}{r}
 \text{no carry generation} \\
 \hline
 0101 \\
 \hline
 \end{array}
 = - (2^{\text{'s}} \text{ complement of } 0101)$$

Result is -ve

$$\begin{array}{r}
 = - (1011) \\
 = - (11)_{10}
 \end{array}$$

• 2's Complement Representation:

$$A = 1011 = -5 \xrightarrow{2's} (0101)_2 = -A$$

$$B = 1110 = -2 \xrightarrow{2's} (0010)_2 = -B$$

$$A - B \rightarrow A + (-B)$$

$$\begin{array}{r} 1011 \\ + 0010 \\ \hline 1101 \end{array} \rightarrow \text{Result} = (1101)_2 = (-3)_{10}$$

$$B - A \rightarrow \begin{array}{r} 1110 \\ + 0101 \\ \hline \end{array}$$

$$\begin{array}{r} 0011 \\ \hline \end{array} \xrightarrow{\text{discard}} \text{Result} = (0011)_2 = (+3)$$

• $A = 1011 = -5$

$$B = 0011 = +3 \xrightarrow{2's} 1101 = -B$$

$$A - B \rightarrow \begin{array}{r} 1011 \\ 1101 \\ \hline \end{array}$$

$$\begin{array}{r} 1011 \\ + 1101 \\ \hline 1000 \end{array} \xrightarrow{\text{discard}} \text{Result} = 1000 = (-8)_{10}$$

• $A = 0101 = +5$

$$B = 1110 = -2 \xrightarrow{2's} (0010)_2 = -B$$

$$(A - B) \rightarrow \begin{array}{r} 0101 \\ 0010 \\ \hline \end{array}$$

$$\begin{array}{r} 0111 \\ \hline \end{array} \xrightarrow{\text{discard}} \text{Result} = (0111)_2 = (+7)_{10}$$



• $n = 4 \text{ bit} \longrightarrow -8 \text{ to } 7 \longrightarrow \text{in case of overflow}$
 $-16, +14$

$n = 5 \text{ bit} \longrightarrow -(16) \text{ to } (15)$
2's complement representation

• $n = 4 \text{ bit} \longrightarrow -7 \text{ to } 7 \longrightarrow \text{in case of overflow}$
 $-14, 14$

$n = 5 \text{ bit} \longrightarrow -(15) \text{ to } 15$

} sign magnitude & 1's complement representation

to avoid overflow in case of addition of two n -bit numbers, we must require $(n+1)$ bits for representation of result.



Topic : 2 Min Summary

→ Signed no. representation.

Thank you

GW
Soldiers !

